

Redefining the Evolutionary History of the Rock Dove, *Columba livia*, Using Whole Genome Sequences

Germán Hernández-Alonso ^{1,2,*†} Jazmín Ramos-Madrigal,^{1,2} Hein van Grouw,³ Marta Maria Ciucani,¹ Emily Louisa Cavill,^{1,2} Mikkel-Holger S. Sinding,⁴ Shyam Gopalakrishnan,^{1,2,5} George Pacheco ¹ and M. Thomas P. Gilbert ^{1,2,6,*}

¹Section for Hologenomics, The Globe Institute, University of Copenhagen, Copenhagen, Denmark

²Center for Evolutionary Hologenomics, The Globe Institute, University of Copenhagen, Copenhagen, Denmark

³Bird Group, Department of Life Sciences, Natural History Museum, Tring, United Kingdom

⁴Department of Biology, University of Copenhagen, Copenhagen, Denmark

⁵Bioinformatics, Department of Health Technology, Technical University of Denmark, Lyngby, Denmark

⁶University Museum, Norwegian University of Science and Technology, Trondheim, Norway

[†]Lead Contact.

*Corresponding authors: E-mails: german_halonso@comunidad.unam.mx; tgilbert@sund.ku.dk.

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Abstract

The domestic pigeon's exceptional phenotypic diversity was key in developing Darwin's Theory of Evolution and establishing the concept of artificial selection. However, unlike its domestic counterpart, its wild progenitor, the rock dove *Columba livia* has received considerably less attention. Therefore, questions regarding its domestication, evolution, taxonomy, and conservation status remain unresolved. We generated whole-genome sequencing data from 65 historical rock doves that represent all currently recognized subspecies and span the species' original geographic distribution. Our dataset includes 3 specimens from Darwin's collection, and the type specimens of 5 different taxa. We characterized their population structure, genomic diversity, and gene-flow patterns. Our results show the West African subspecies *C. l. gymnocyclus* is basal to rock doves and domestic pigeons, and suggests gene-flow between the rock dove's sister species *C. rupestris*, and the ancestor of rock doves after its split from West African populations. These genomes allowed us to propose a model for the evolution of the rock dove in light of the refugia theory. We propose that rock dove genetic diversity and introgression patterns derive from a history of allopatric cycles and dispersion waves during the Quaternary glacial and interglacial periods. To explore the rock dove domestication history, we combined our new dataset with available genomes from domestic pigeons. Our results point to at least 1 domestication event in the Levant that gave rise to all domestic breeds analysed in this study. Finally, we propose a species-level taxonomic arrangement to reflect the evolutionary history of the West African rock dove populations.

Key words: *Columba livia*, domestication, admixture, refugia theory, population genomics.

Introduction

The pigeon (*Columba livia*) is one the most well-known birds worldwide due to the near ubiquitous distribution of its feral populations, and the extended breeding practices of its domestic form. The domestic pigeon, which on occasion has been bred to extreme exuberance, has been an important model organism for the study of evolution, behavior, and genotype expression, among other research areas (Helms and Bruggmann 2007). In contrast to the domestic pigeon, its parental species, the rock dove, has received relatively little attention and consequently, several unresolved questions remain. For instance, the intraspecific taxonomic classification of the rock dove is characterized by significant ambiguity, with several inconclusive or invalidated subspecies. The genetic diversity of

the rock dove remains largely unexplored, including its evolutionary history, ongoing evolutionary dynamics, and the extent of its admixture with feral pigeons (free-living birds mainly descended from domestics). Consequently, it is difficult to determine their conservation status, as feral pigeons may have replaced wild rock dove populations in several locations. Furthermore, the number of rock dove domestication events and their geographic locations remain uncertain.

Archaeological evidence suggests that the rock dove was first domesticated between 3,000 to 10,000 yr ago in either the Mediterranean Basin (Johnston 1992; Johnston and Janiga 1995; Stringham et al. 2012) or in the Fertile Crescent alongside the Neolithic Revolution (Driscoll et al. 2009). It has been argued that rock doves were probably domesticated several times and in different places

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